REG.NO.:

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering and Information Systems

CONTINUOUS ASSESSMENT TEST - I

WINTER SEMESTER 2024-2025

SLOT: C1+TC1

Programme Name & Branch	: MCA
Course Code and Course Name	: Full Stack Web Development & PMCA601L
Faculty Name(s)	: Prof. Sumangali K, Prof. B. Senthil Murugan
Class Number(s)	: VL2024250503350, VL2024250503346
Date of Examination	: 29-1-2025
Exam Duration	: 90 minutes

Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes (Type the CO statements covered in this question paper. Use the CO number as per the syllabus copy)
  CO1- Develop interactive and responsive web pages using HTML, CSS and Bootstrap
  CO2- Use JavaScript and JQuery to create dynamic web pages

Q. No	Question	M	CO	BL
1.	Create an HTML form layout for an employment information system where employees or job applicants can input and submit their personal and professional information details. The form should contain details such as personal details (first name, last name, DOB, phone number, and email address), Job Preferences (with a dropdown menu), Work Experience (with range of numbers minimum 3 and maximum 10 years) and List of Skills (with checkbox). Apply the following CSS styling specifications for the form elements as mentioned below: a)Use Autofocus b)Placeholder's & Mandatory fields c) Background color	10	1	3
2.	XYZ Bank is a leading financial institution in the city, offering various services such as savings accounts, loans, credit cards, and investments. The bank has multiple departments like Retail Banking, Corporate Banking, and Wealth Management. Customers can open accounts, apply for loans, or avail of other banking services through branch visits or online platforms. When visiting the bank for the first time, customers' details are recorded, and a unique Customer ID is generated. For future transactions or service requests, customers need to provide their Customer ID. Design a suitable application architecture for XYZ Bank's system automation. Enumerate the system features and briefly explain the technology and software toolkit used for each application layer.	10	1	2
3.	Design a web page using **HTML**, **CSS**, and **JavaScript** that validates credit card numbers based on the following requirements. The validation should use	10	2	4

	regular expressions to ensure the credit card numbers adhere to the specific formats of major credit card types. i) The page should include a dropdown box to select the credit card type (e.g., Diners Club, Visa, MasterCard, American Express). ii) Provide a text input field to allow the user to enter the credit card number. iii) Include a "Validate" button that, when clicked, checks whether the entered credit card number is valid. iv) Display a message below the button indicating whether the credit card number is valid or invalid. Major credit card numbers must be in the following formats and display the possible validation error messages if required. American Express—Numbers start with 34 or 37 and consist of 15 digits. Diners Club—Numbers begin with 300 through 305, or 36 and 38 and consists of 14 digits Discover—Numbers begin with 6011 or 65 and consist of 16 digits. JCB—Numbers beginning with 2131 or 1800 consist of 15 digits, while numbers beginning with 35 consist of 16 digits. MasterCard—Numbers start with the numbers 51 through 55 and consist of 16 digits. **Sample Output:** **Validate Credit Cards** Credit card: Diners Club Number: 123 Validate Invalid credit card number			
4.	Write a HTML code to design a form which collects the input string for an array, few buttons like 'Add', 'Delete' and 'Print'. Create a JavaScript functions for the following: a.Add button should insert the given string into the array if it is not null b.Delete button should remove the value if it is existing in the array. Also, inform the user about the process whether deleted or the input is not available. c.Print button should display the entire content of the array in the HTML page.	10	2	3

	Sample Output: Enter a String: Chikku ADD DELETE PRINT **Array Contents** Apple Banana Kiwi Chikku			
5.	Develop a Java Script application that takes multiple lines of input from the user and processes the data as follows: i) Display the unique sentences entered by the user. ii) Count and display the number of occurrences of each unique word in the text. For example, if the input text describes a shopping list like: *"Buy apples and bananas. Apples are fresh."* The unique sentences are: *"Buy apples and bananas."* and *"Apples are fresh."* The word occurrences are: *"apples" occurs 2 times*, *"and" occurs 1 time*, and so on. Apply distinct styles for displaying the unique sentences and word occurrences to enhance readability.	10	2	3

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